

The Algebra Masterclass

1. Vieta's Formulas

The Idea

This is the most important “secret code” in algebra. It tells you the sum and product of the roots of a polynomial just by looking at its coefficients. You don't need to find the roots at all! For a quadratic equation $ax^2 + bx + c = 0$ with roots α and β :

$$\text{Sum of roots: } \alpha + \beta = -b/a$$

$$\text{Product of roots: } \alpha\beta = c/a$$

When to Use It

Use this whenever a problem gives you an equation and asks for the value of something like $\alpha^2 + \beta^2$, $(\alpha - \beta)^2$, or $\frac{1}{\alpha} + \frac{1}{\beta}$. The trick is to rewrite the target expression using only $(\alpha + \beta)$ and $(\alpha\beta)$.

Practice Problem 1 (1). *The roots of the quadratic equation $x^2 - 6x + 2 = 0$ are α and β . What is the value of $\alpha^2 + \beta^2$?*

Hint: Think about the expression $(\alpha + \beta)^2$. When you expand it, what do you get? How can you adjust it to get just $\alpha^2 + \beta^2$?

2. Clever Substitution

The Idea

Sometimes an equation looks terrifyingly long and complicated, but it has a repeating part. By giving this repeating part a simple nickname (like replacing $x^2 + 5x$ with u), the monster equation often turns into a simple one you already know how to solve.

When to Use It

Look for any chunk of algebra, big or small, that appears more than once in an equation. This is your signal to substitute!

Practice Problem 2 (2). Find all real numbers x that solve the equation: $(x^2 + 2x - 1)^2 - 9(x^2 + 2x) + 23 = 0$.

Hint: The expression $x^2 + 2x$ is almost perfectly repeated. Let $u = x^2 + 2x$. How can you rewrite the equation using u ?

3. Symmetric Systems

The Idea

Some systems of equations are like a perfect partnership. If you swap x and y , the equations don't change at all. The fastest way to solve them is to find their sum ($s = x + y$) and their product ($p = xy$) first.

When to Use It

When you have two equations, and you can see that x and y are interchangeable. Once you find the values for s and p , you can build a new quadratic, $t^2 - st + p = 0$, whose roots will be your answers for x and y .

Practice Problem 3 (3). Find all real pairs (x, y) that satisfy the system: $x + y + xy = 17$ and $xy(x + y) = 70$.

Hint: Let $s = x + y$ and $p = xy$. Rewrite the equations as $s + p = 17$ and $ps = 70$. Solve for s and p , then find x and y as roots of $t^2 - st + p = 0$.

4. Reciprocal Equations

The Idea

A special type of polynomial where the coefficients read the same forwards and backwards (e.g., $1, -5, 8, -5, 1$). These have a beautiful weakness. If you divide the whole equation by its middle power of x , it folds up perfectly.

When to Use It

Check the coefficients. If they form a palindrome, this trick is almost guaranteed to work.

Practice Problem 4 (4). Solve the equation $x^4 - 5x^3 + 8x^2 - 5x + 1 = 0$.

Hint: Since $x = 0$ is not a solution, you can safely divide the entire equation by x^2 . Then, group the terms that look alike, like $(x^2 + \frac{1}{x^2})$ and $(x + \frac{1}{x})$. Use a substitution for the simpler group.

5. Completing the Square

The Idea

You know “completing the square” as a way to solve quadratics. But its real power is in changing the *form* of an expression to reveal its properties, like its minimum or maximum value.

When to Use It

Whenever a problem asks for the minimum or maximum value of a quadratic-like expression, especially one with multiple variables.

Practice Problem 5 (5). Find the minimum possible value of the expression $A = x^2 + y^2 + 4x - 10y + 35$ for any real numbers x and y .

Hint: Treat the x terms and y terms separately. Complete the square for $x^2 + 4x$. Then, complete the square for $y^2 - 10y$. See what you have left.

6. Simon’s Favorite Factoring Trick

The Idea

A clever method for equations where you need to find integer solutions. The goal is to add a specific number to both sides of the equation so that one side magically factors into a product of two binomials.

When to Use It

When you have an equation with integer solutions that involves a term like xy along with x and y terms (e.g., $xy - 3x - 5y = 10$).

Practice Problem 6 (6). Find all pairs of positive integers (x, y) that satisfy the equation: $\frac{1}{x} + \frac{1}{y} = \frac{1}{4}$.

Hint: First, get rid of the fractions by multiplying the entire equation by $4xy$. This will give you an equation like the one described above. Then, try to force it to factor into the form $(x - a)(y - b) = c$.

7. Telescoping Sums

The Idea

A technique for calculating long sums that look intimidating. The secret is to rewrite each term in the sum as a *difference* of two things. When you write out the whole sum, the second part of each term cancels out the first part of the next term, leaving only the very first and very last pieces.

When to Use It

When you see a sum of fractions in a clear pattern. The most famous pattern is $\frac{1}{n(n+1)}$.

Practice Problem 7 (7). Calculate the exact value of the sum: $S = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \cdots + \frac{1}{99 \cdot 100}$.

Hint: The key is the identity $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$. Try rewriting the first few terms of the sum using this rule and watch the terms cancel out.

8. Difference of Squares

The Idea

The simple identity $a^2 - b^2 = (a - b)(a + b)$ is one of the most powerful factoring tools in existence. It can be used in surprising ways to simplify what look like very difficult calculations.

When to Use It

Look for expressions that are almost a difference of squares, or chains of products where this pattern can be started with a clever trick.

Practice Problem 8 (8). Find the value of the product: $P = (3+1)(3^2+1)(3^4+1)(3^8+1)(3^{16}+1)$.

Hint: This chain of terms is just asking to be part of a telescoping product. What if you multiplied the whole expression by $(3-1)$? Remember to divide by it at the end to keep the value the same.

9. Infinite Nested Radicals

The Idea

Expressions with square roots inside square roots that go on forever look infinitely complex, but they often have a very simple, finite value. The trick is to recognize that the expression contains a copy of itself.

When to Use It

When you see the tell-tale “...” inside a repeating radical expression.

Practice Problem 9 (9). Find the exact value of $x = \sqrt{6 + \sqrt{6 + \sqrt{6 + \dots}}}$.

Hint: Let the entire expression be equal to x . Now look at the part of the expression under the very first square root sign. What is it equal to? This should give you a very simple quadratic equation to solve for x .

10. Sophie Germain's Identity

The Idea

A beautiful and non-obvious factorization trick that shows how to factor expressions of the form $a^4 + 4b^4$. The key is to add and subtract a middle term to create a perfect square, which then leads to a difference of squares. The most famous case is $x^4 + 4 = (x^2 + 2x + 2)(x^2 - 2x + 2)$.

When to Use It

When you encounter a polynomial that seems unfactorable, like a sum of two fourth powers, this creative technique might be the answer.

Practice Problem 10 (10). Find the value of the expression $\frac{2024^4 + 4}{2023^2 + 1}$.

Hint: The numerator is in the form $a^4 + 4b^4$ if you let $a = 2024$ and $b = 1$. Apply the identity to factor it. The denominator might simplify if you expand the squares. Look for cancellations!

11. Systems by Creative Combination

The Idea

Not all systems of equations are symmetric. Sometimes the key is not substitution, but adding or subtracting the equations from each other to produce a new, much simpler equation.

When to Use It

When direct substitution looks like it will create high-degree polynomials and a huge mess. Always check if adding or subtracting the equations reveals a hidden structure.

Practice Problem 11 (11). *Given the equations $x^2 + xy + x = 10$ and $y^2 + xy + y = 20$, find all pairs of real numbers (x, y) that are solutions.*

12. Transformation of Roots

The Idea

An elegant technique to create a new polynomial whose roots are a function of the roots of an old one (e.g., shifted, squared, or inverted), without ever finding the original roots.

When to Use It

When a problem gives you a polynomial and asks you to build another polynomial whose roots are related to the original roots in a simple, uniform way.

Practice Problem 12 (12). *The roots of the equation $x^2 - 10x + 5 = 0$ are α and β . Find a new quadratic equation whose roots are $\alpha + 2$ and $\beta + 2$.*

Hint: Let the new root be y . So, the relationship is $y = x + 2$. Rearrange this to express the old root in terms of the new one: $x = y - 2$. Now, substitute this expression for x back into the original equation and simplify. The result will be a new equation for y with the desired roots.